Hybrid robot control with distance fields

Use GPT for local task intent. Use a local planner for repeated motion.

Compact scene description → graph distance-to-go → measured-pose tracking, with Direct control near contact.

Current Dijkstra version · Astra medium · ChatGPT subscription
Source and recorded experiments reviewed on October 12, 2026



Real obstacle-scene observation.
No synthetic scene or trajectory is used in the data figures.

Scope: current distance-field implementation; historical comparisons retain their original evaluation criteria.

OVERVIEW

The division of work

The acceleration hypothesis is fewer model calls during free-space motion.

Observe

Task + two RGB views
Measured robot pose
Observable history

→

GPT decision

Route + local goal
Predicted obstacles
Orientation + gripper

→

Local planning

Build valid grid
Solve distance-to-go
Extract one path

→

Robot control

Track measured pose
Execute bounded action
Read new observation

Local feedback continues every action. A model refresh occurs at a controller boundary or an observable event.

[M1–M4] distance_subscription.py, distance_policy.py, distance_planner.py, tracking.py; current code, not a learned rollout model.

POLICY INTERFACE

What the model sees—and what it predicts

Zero-shot means no demonstration episode; common hand-written task guidance is still present.

Input

Task instruction
External camera + wrist RGB
8D robot state: end-effector xyz, axis-angle, two finger joints
Recent live images + summarized action feedback

Optional one-demo prefix only in marked experiments.

Output

Field branch: compact geometry/cost parameters, pose/gripper intent and control context.

Direct branch: five normalized 7D actions.

No ground-truth object pose, depth, contact or success signal enters the model.

[M1,M5] distance_subscription.py, zero_shot_direct_subscription.py, observation.py; latest two live image pairs retained.

A real compact Field output

Snapshot A · zero-shot obstacle Hybrid · decision before action index 201 (indices are zero-based).

goal_offset_m	[-0.070, 0.000, +0.045]
attraction_weights	[1, 1, 1]
repulsors	[]
blocked lower	[+0.015, -0.080, -0.080]
blocked upper	[+0.080, +0.080, -0.015]
unknown lower	[-0.080, -0.080, -0.080]
unknown upper	[+0.015, +0.080, -0.050]
gripper / phase	open / recover

Small structured output
Goal, up to four Gaussian repulsors, up to four blocked boxes and four unknown boxes.

All offsets are in meters along world axes, relative to the end-effector position when this Field was generated.

GPT does not emit 35,937 scalar distances.

[S1] trace.jsonl action 201 / effective_compact_field; raw snapshot saved in deck_assets/snapshot_201.json.

Expand locally into a bounded planning domain

The grid is centered once per Field refresh. Reuse does not expand known space.

33 × 33 × 33

35,937 grid vertices

5 mm

Spacing along each grid axis

16 cm

Span per axis: ±8 cm about the center

Coordinates: $x_{ijk} = c + 0.005 \cdot (i-16, j-16, k-16)$

Validity: inside workspace AND outside predicted blocked / unknown regions.
Check graph edges and tracking chords against masks and closed boxes.

Unknown is not free.

Repulsors add soft cost; they do not create collision geometry.

5 mm resolution is a sampling choice, not 5 mm perception accuracy.

[M2,M3,M6] compact_field.py:118; distance_planner.py:67,128; schema.py GridSpec. Grid/mask geometry is model-predicted.

Definition: graph distance-to-go

Distance is to a local goal region through predicted valid space—not distance to an obstacle surface.

TRAVERSAL DENSITY

$$\rho(x) = 1 + \Sigma_k \beta_k \exp(-\|x-r_k\|^2 / (2\sigma_k^2))$$

COST OF A VALID EDGE

$$w(u,v) = \|x_v-x_u\| \cdot [\rho(x_u)+\rho(x_v)] / 2$$

DISTANCE-TO-GO

$$D(v) = \min_{P:\, v\to G} \Sigma_{(u,z)\in P} w(u,z)$$

26-neighbor graph
Reverse multi-source Dijkstra stores D and a successor toward the goal.

Units
Meters with no repulsors; weighted meters with repulsors.

Invalid / unreachable: +∞.
Goal endpoints: D = 0.

[M3] G: valid grid/virtual goal-region endpoints; terminal-edge costs included. Enclosing corners can have D > 0. Tolerances: slide 9.

DEFINITION

Three fields that should not be conflated

The current path objective is D. The compatibility potential U is still stored but does not drive Dijkstra.

Quantity	Meaning	Role / limitation
Legacy potential U	Weighted goal attraction + Gaussian bumps; dimensionless.	Expanded into GridField for compatibility. Current Dijkstra ignores U and attraction_weights.
Current distance D	Minimum weighted route length to the local goal region.	Computed locally on the valid graph. Exact for that finite augmented graph; not exact continuous geodesic distance.
Conventional SDF	Signed distance to the nearest obstacle surface.	Different quantity. This implementation does not reconstruct a surface-distance field from sensor geometry.

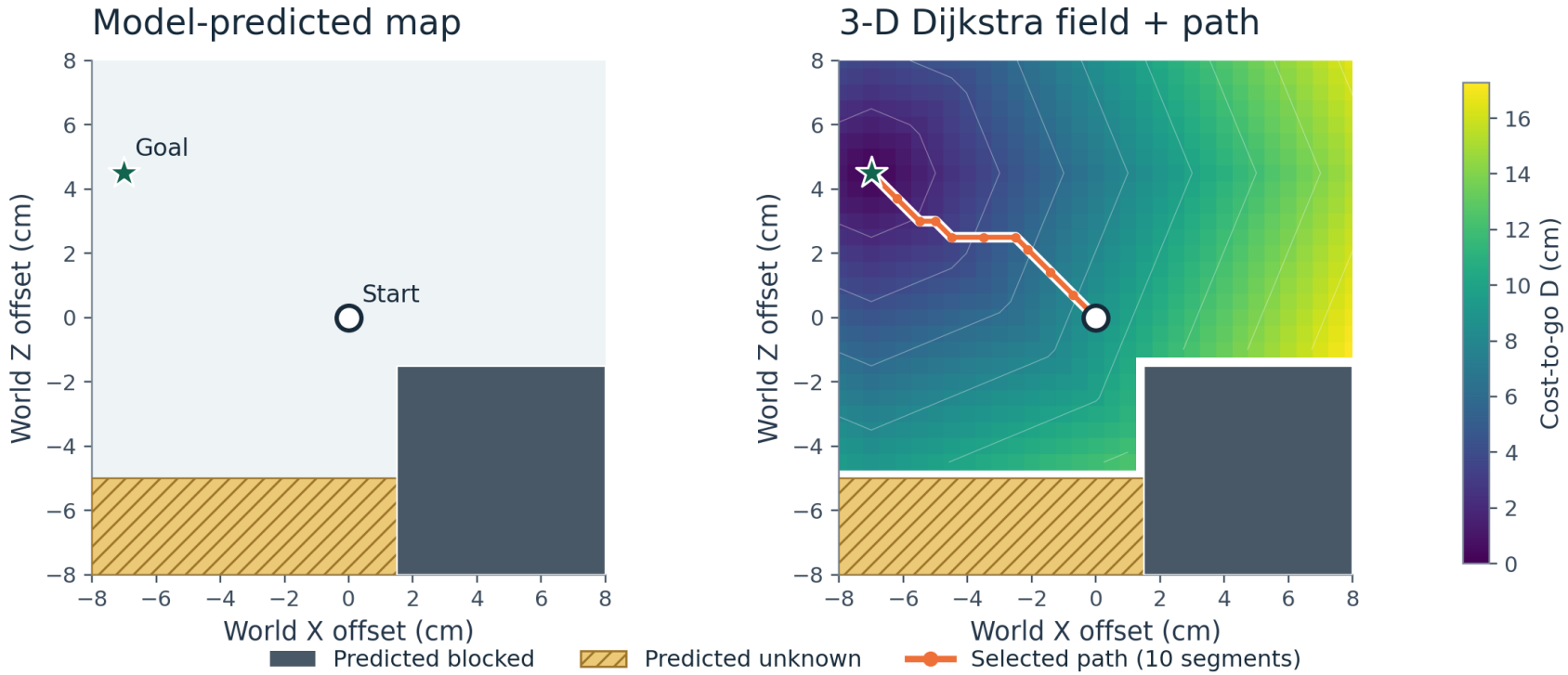
No claim of a full-scene metric, a Riemannian neural field, or a certified robot configuration-space planner.

[M2,M3] compact_field.py:118; distance_planner.py:28,228. Definitions distinguish representation, objective and geometry.

SNAPSHOT A

Real data: predicted masks and solved distance

A slice of the full 3D solve at the recorded start Y; the selected path lies in this XZ plane.



Actual XZ slice at world Y = 0.20656 m; selected path lies in-plane. Masks are model estimates, not object meshes.

RECORDED FIELD 201

8.864 cm

Graph cost at the start

33^3 nodes · 5 mm spacing

26-neighbor 3-D search

Goal radius: 2.5 mm

ρ = 1; no repulsors

White: invalid / unreachable

Snapshot A has no Gaussian repulsors: ρ = 1. The plot shows the model's local masks and graph distances, not recovered physical obstacle geometry.

[S1] zero-shot Hybrid action 201. Frozen-code reconstruction; axes/units and exact numerical arrays are in field_assets/.

PATH EXTRACTION

Planning returns one route, not a trajectory bank

35,937 possible grid vertices are not 35,937 candidate trajectories.

1

Connect the measured start

Validate connections to enclosing grid corners; choose start-edge cost + D.

2

Follow stored successors

Extract a shortest graph route. Reuse cached D while the field and goal are unchanged.

3

Keep a short executable prefix

Merge only collinear edges. At most 10 segments, each ≤1 cm; at most 10 cm of geometric path.

Current tolerances: 2.5 mm goal-region radius + 2.5 mm tracking endpoint tolerance; outer arrival check = 5 mm and 0.05 rad. Invalid/out-of-map goals request a new observation.

[M3,M4] distance_planner.py:262–337; distance_policy.py:28–35,104,423. Success: one CandidateRecord; hold/failure: zero.

EXECUTION

Convert the geometric path into robot actions

Geometric segments, control ticks and model calls are different quantities.

Measured-pose path tracking

Project actual end-effector position onto the remaining path.

Choose a checked point up to **3 cm ahead**.

Command position error, bounded to **5 cm norm**. Shorten the reference if its chord crosses a predicted mask.

$$\delta x = \text{bounded}(x_{\text{reference}} - x_{\text{measured}})$$

$$a = [\delta x / 0.05, \delta\theta / 0.5, \text{gripper}]$$

Rotation error is capped at 0.15 rad.
Gripper: open −1, close +1, hold previous.
Actions go to the 20 Hz OSC controller.

Track for at most 10 control ticks per local plan, or replan earlier. A “10-step path” does not guarantee 10 cm of actual robot travel.

[M4] tracking.py:63–145; action.py:40–96; distance_policy.py:104. Commands are bounded setpoint increments, not measured displacements.

CLOSED LOOP

How stages and controller switches work

GPT selects phase and branch. Typical uses are shown below; phase names do not impose a fixed controller mapping.

Field branch

Approach / transport / alignment.
Local path tracking; initially up to **30 actions** per model response.

Reliable visual progress may extend reuse to 60, 90, then 120 actions.

Direct branch

Grasp / release / contact recovery.
Five model-generated actions per response, executed unchanged.

Task termination can interrupt the final chunk.

Refresh early: measured arrival, invalid path/reference, boundary, low realized motion, lost tracking, or placement verification concerns. A phase label or “near_target” prediction is not proof of contact or success.

[M4,M5] distance_policy.py:218,340,426; hybrid_policy.py:149,193. Cadence is event-driven; 30 × 1-step replans is not the current loop.

LOCAL FEEDBACK

Ground the goal and correct it when evidence allows

These mechanisms are conditional. Their existence in code does not mean they activated in a rollout.

Account for the held bowl

$$g = c + d - R_{target} R_{current}^{T} r$$

c: Field center; d: desired bowl offset.
r: current bowl offset from the gripper.

Subtract the rotated grasp offset from the bowl goal. Assumes a visually estimated, rigid grasp.

Optional image-based XY correction

Track model-specified image points.
Fit a local 2×2 image Jacobian from measured XY motion.
Apply a validated correction ≤1 cm.

No depth reconstruction; masks and original grid center stay fixed.

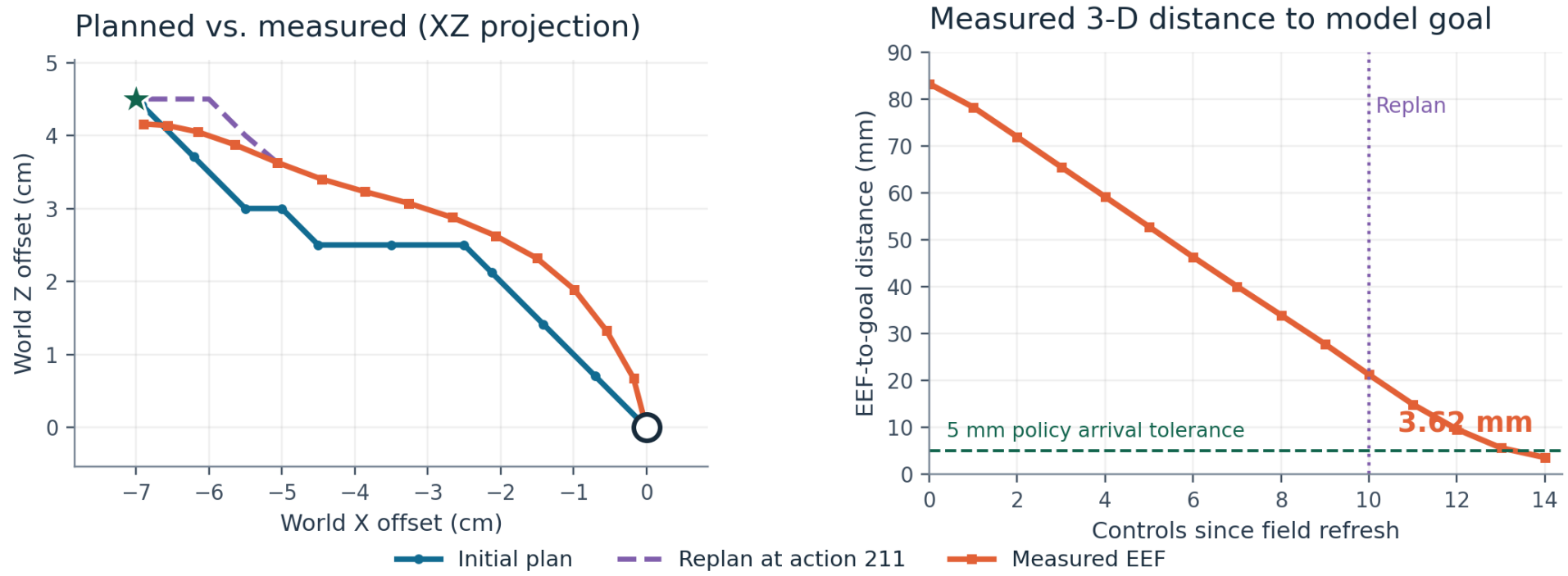
Latest zero-shot run: all 13 Field outputs had no visual anchor; all 310 feedback records reported anchor unavailable. No local visual goal corrections or reuse extensions occurred.

[M4,M7,S2] field_grounding.py:91; local_visual_feedback.py:294; distance_policy.py:297. Visual agreement never certifies task success.

SNAPSHOT A EXECUTION

Real data: a plan is not the measured motion

Recorded actions 201–214, including local replanning; no simulated trajectory was substituted.



Recorded actions 201–214; 14 controls, two local plans. Positions are relative to the fixed field-generation center.

A 10-segment geometric prefix can be shorter than 10 cm. Tracking uses feedback to reduce error; commanded displacement is not equal to realized displacement.

[S1] planned paths and measured end-effector states, trace.jsonl. Reconstruction metadata and CSV/NPZ are saved with the figure.

MEASURED RESULTS

Evaluation protocol and the timing metric

Do not mix different success criteria or count failed-attempt durations as completed-task speedups.

Common condition	Setting
Model / channel	gpt-6-astra · medium · ChatGPT subscription OAuth · zero API-key calls
Within each paired condition	Same task, initial state/scene, demo if present, shared guidance and observable feedback
Timing	T = recorded control_elapsed_s − video_render_s
Earlier completed pairs	Original LIBERO goal only; release/stability and collision freedom were not separately certified
Latest zero-shot obstacles	LIBERO goal AND no forbidden contact; immediate failure at first forbidden physics contact

One trial per arm and condition. Video subtraction is retrospective accounting, not a recording-disabled rerun.

[R1–R5] frozen summaries/manifests/audits. Timings include model waiting, local control, simulator/observation and loop overhead.

GOAL-ONLY RESULTS

Earlier matched pairs reached the LIBERO goal

One demonstration per method. These runs did not use the later collision-free success criterion.

Task / condition	Hybrid	Direct	Less time	Calls H / D
Drawer + bowl	13:57	16:37	16.1%	63 / 85
Bowl: original layout	2:57	3:58	25.6%	15 / 21
Bowl: relocated objects	3:36	3:40	1.9%	17 / 19

Times are min:sec after removing measured video overhead. Hybrid used fewer model requests in all three pairs; the gain ranged from 1.9% to 25.6%.

Single trials—not a reliability estimate. The 1.9% result is small; raw recorded control time actually favored Direct before video subtraction.

[R1–R3] results_notes.json, paired_conditions. Both arms stopped on the original goal predicate; post-release stability not verified.

BOWL / ORIGINAL LAYOUT / ONE DEMO

A completed bowl pair illustrates the mechanism

Less model interaction, even with more executed actions. This is one observed pair.

HYBRID

177.0 s

15 requests · 131 actions
93 Field actions / 38 Direct actions
Model/client pipeline: 166.3 s



Model/client pipeline: 94.0% of adjusted time

DIRECT

237.8 s

21 requests · 101 actions
0 Field actions / 101 Direct actions
Model/client pipeline: 232.1 s



Model/client pipeline: 97.6% of adjusted time

Pipeline time includes client/transport work and local expansion. It is not pure GPU inference latency; timer components must not be added blindly.

[R2] bowl_distance_medium_one_demo_001 and bowl_direct_medium_one_demo_001 / summary.json; 25.6% adjusted time reduction.

ONE-DEMO OBSTACLE RUNS

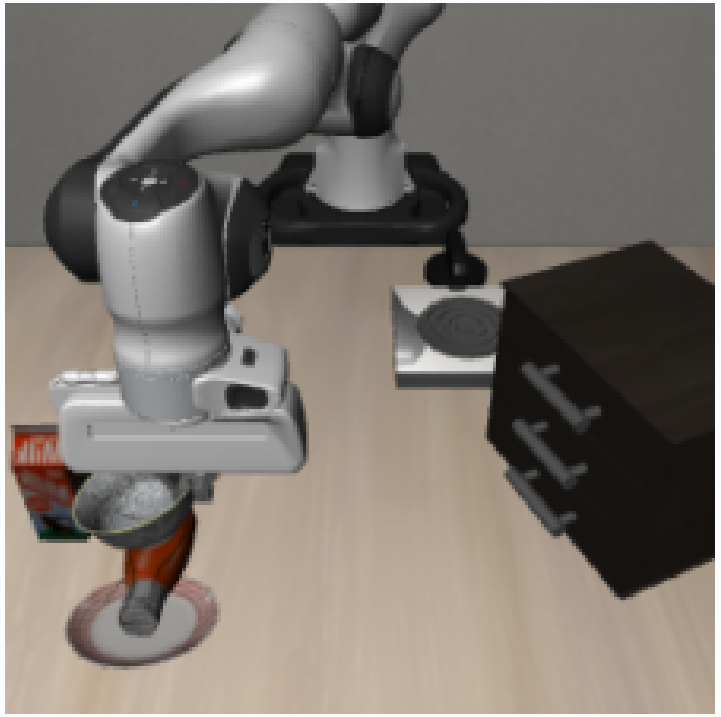
Obstacle tasks expose a different failure mode

Both reached the original placement predicate, but both collided with the ketchup bottle: both are failures.

Hybrid



Direct



Adjusted attempt duration
Hybrid: 6:41
Direct: 8:50

No valid task speedup.
A goal-only success flag did not enforce obstacle preservation.

Review corrected both outcomes to collision failure.

Recorded RGB excerpts show the bottle displaced/tilted. Earlier Hybrid: all eight Field responses had empty obstacle/unknown/repulsor lists.

[R4] corrected comparison v2 + collision_evidence/. Images are real recorded frames; these times are not times to first contact.

STRICTER OBSTACLE EVALUATION

Latest zero-shot test: no successful pair

Same frozen obstacle scene; explicit no-collision guidance; actual physics-substep contact monitoring.

Method	Outcome	Time*	Actions	Requests
Hybrid	Collision failure: left finger–ramekin	9:52	310	44
Direct	User-stopped; no goal reached	25:07	505	102†

* Video-excluded attempt duration, not completion time. † 101 valid responses + one interrupted request.



Hybrid / collision stop



Direct / user stop

Neither reached LIBERO success. Direct remained incomplete and unscored, with no forbidden contact before the user stop.

Allowed: finger–target bowl grasp; bowl–table/plate support.
Excluded: robot self/internal and scene-scene-only contact.
Monitoring detects contact; it does not prevent first contact.

[R5,S2] comparison.json, task_evaluation.json and collision.json. Hybrid stopped at zero-based action 309, physics substep 7.

INTERPRETATION

What the evidence supports—and what is missing

Fewer model requests do not by themselves establish successful or collision-free execution.

Supported by these recordings

Fewer model requests in three earlier completed goal-only pairs.

Compact outputs avoid writing a dense 35,937-value response.

Current local planning is a well-defined shortest-path computation over predicted valid space.

Not demonstrated yet

Reliable collision-free manipulation in the harder scene.

Correct target identification from zero-shot RGB.

Whole-arm / payload swept-volume safety, or consistent activation of local visual correction.

Priority: verifiable target localization → usable visual anchors → gripper/payload geometry checks → repeated matched trials.

[M1–M7,R1–R5,S2] These are implementation properties and single-run observations, not generalized performance guarantees.

TECHNICAL DETAIL

Appendix: exact potential and distance semantics

The code retains the legacy potential representation alongside the new distance objective.

$$U(x) = \sqrt{\left[\Sigma_a\, \alpha_a((x_a - g_a)/\ell)^2\right]} + \Sigma_k\, \beta_k \exp(-\|x - r_k\|^2 / (2\sigma_k^2))$$

ℓ = 0.08 m. U is dimensionless. Attraction weights α affect U, but not current Dijkstra edge weights.

- **Distance is goal-conditioned**

  D is a cost-to-go to one terminal region. It is not a scalar clearance to nearby obstacle surfaces.

- **Optimality has a precise scope**

  Reverse Dijkstra is exact on the constructed finite graph with endpoint-averaged density and valid terminal connections. It is not a continuous Eikonal/PDE solve.

[M2,M3] compact_field.py:118–146; distance_planner.py:128–201. A Gaussian cost penalty alone never creates a hard obstacle.

RESULTS LEDGER

Appendix: all paired conditions at a glance

Same current Dijkstra family; criterion and demonstration count are explicit for every row.

Condition	Demos	Outcome H / D	H seconds*	D seconds*
Drawer	1	Goal / goal	836.7	997.3
Bowl / original	1	Goal / goal	177.0	237.8
Bowl / relocated	1	Goal / goal	215.8	220.0
Bowl / obstacles	1	Collision / collision	400.5	529.6
Bowl / obstacles	0	Collision / incomplete	592.1	1506.7

* Video-excluded time. First three rows: earlier goal-only completion. Last two rows: failed/incomplete attempts, excluded from completion-speed comparisons. No pooled average is meaningful across these criteria.

[R1–R5] Exact timing subtraction, raw timings, request/actions and source hashes are included in results_notes.json.

EVIDENCE

Appendix: source and reproducibility map

All visuals and numbers are derived from local source or recorded experiment artifacts. No new robot test was run for this deck.

Reference	Where to verify
M1–M7	src/gpt_field/: distance_subscription.py, compact_field.py, distance_planner.py, distance_policy.py, tracking.py, action.py, field_grounding.py, local_visual_feedback.py
R1–R3	runs/{drawer_one_demo, bowl_one_demo, bowl_relayout_one_demo}_matched_comparison_001/ and paired raw summaries
R4	runs/bowl_obstacle_one_demo_matched_comparison_001/comparison.json — corrected collision-aware report
R5 / S2	runs/bowl_obstacle_zero_shot_matched_comparison_001/{comparison.json, analysis.json, evidence/}
S1	runs/bowl_obstacle_distance_medium_zero_shot_001/trace.jsonl — snapshot 201 and measured execution

Delivered with build source, numerical field/trajectory assets, provenance hashes, method_notes.json and results_notes.json.